Using Laplace Transform to solve PDEs

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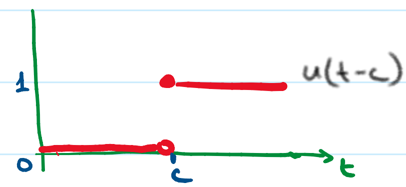
10- Use Laplace Transform to solve PDEs

10.1 Recall:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = F(s) \qquad \mathcal{L}^{-1}\{F(s)\} = f(t)$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0) \qquad \mathcal{L}^{-1}\{e^{-cs}F(s)\} = \underbrace{u(t-c)}_{\text{Heaviside}}\, f(t-c)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$



10.2 Semi-Infinite String

Find the displacement $w(x, t)$ of an elastic string subject to the following conditions. (We write w since we need u to denote the unit step function.)

(i) The string is initially at rest on the x-axis from $x = 0$ to ∞ (*"semi-infinite string"*).

(ii) For $t > 0$ the left end of the string ($x = 0$) is moved in a given fashion, namely, according to a single sine wave

$$w(0, t) = f(t) = \begin{cases} \sin t & \text{if } 0 \leqq t \leqq 2\pi \\ 0 & \text{otherwise} \end{cases} \qquad \text{(Fig. 316).}$$

(iii) Furthermore, $\lim_{x \to \infty} w(x, t) = 0$ for $t \geqq 0$.

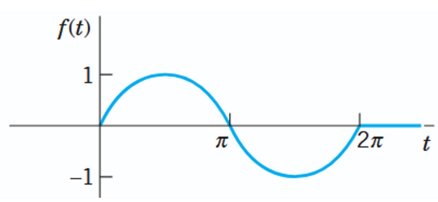


Fig. 316. Motion of the left end of the string in Example 1 as a function of time t

See: battle rope exercise: [How to perform BATTLE ROPES - HOIST Fitness MotionCage Exercise]

10.3 The <u>Problem</u>:

Wave equation: $\frac{\partial^2 \omega}{\partial t^2} = c^2 \frac{\partial^2 \omega}{\partial x^2}$ for $x>0,\ t>0$

<u>BC</u>: $\omega(0,t) = f(t) = \begin{cases} \sin(t), & 0 \le t \le 2\pi \\ 0 & \text{otherwise} \end{cases}$; $\lim_{x\to\infty} \omega(x,t) = 0$

<u>IC</u>: $\omega(x,0) = 0$, $\omega_t(x,0) = 0$

10.4 Using Laplace Transform:

$$\mathcal{L}\left\{\frac{\partial^2 \omega}{\partial t^2}\right\} = s^2 W(x,s) - s\underbrace{\omega(x,0)}_{=0} - \underbrace{\omega_t(x,0)}_{=0} = s^2 W(x,s)$$

$$\mathcal{L}\left\{\frac{\partial^2 \omega}{\partial x^2}\right\} = \frac{\partial^2}{\partial x^2}\mathcal{L}\{\omega\} = \frac{\partial^2}{\partial x^2} W(x,s)$$

Then $\frac{\partial^2 \omega}{\partial t^2} = c^2 \frac{\partial^2 \omega}{\partial x^2}$ becomes: $s^2 W(x,s) = c^2 \frac{\partial^2}{\partial x^2} W(x,s)$

$$\Rightarrow \frac{\partial^2}{\partial x^2} W - \frac{s^2}{c^2} W = 0$$

10.5 Solving the DE: Similar to $W'' - \frac{s^2}{c^2} W = 0 \Rightarrow W = Ae^{sx/c} + Be^{-sx/c}$

$$W(x,s) = A(s)\, e^{sx/c} + B(s)\, e^{-sx/c}$$

<u>Now</u>: $\lim_{x\to\infty} W(x,s) = \lim_{x\to\infty} \int_0^\infty e^{-st}\, \omega(x,t)\,dt = \int_0^\infty e^{-st}\left(\lim_{x\to\infty} \omega(x,t)\right)dt = 0$

Thus: $0 = \lim_{x\to\infty} W(x,s) = \lim_{x\to\infty}\left[A(s)e^{sx/c} + B(s)e^{-sx/c}\right] = A(s)\lim_{x\to\infty} e^{sx/c} + B(s)\underbrace{\lim_{x\to\infty} e^{-sx/c}}_{=0}$

$$\Rightarrow \quad 0 = A(s)\underbrace{\lim_{x\to\infty} e^{xs/c}}_{\neq 0} \Rightarrow A(s) = 0$$

Thus $W(x,s) = B(s)e^{-sx/c}$. Note that $W(0,s) = B(s)$.

Now: $W(0,s) = \int_0^\infty e^{-st}\,\omega(0,t)\,dt = \int_0^\infty e^{-st} f(t)\,dt = F(s)$

Therefore, $W(x,s) = F(s)\,e^{-sx/c}$

10.6 Apply inverse Laplace:

Since $\mathcal{L}^{-1}\{F(s)\,e^{-as}\} = f(t-a)\,u(t-a)$

in this case:

$$\omega(x,t) = \mathcal{L}^{-1}\{W(x,s)\} = \mathcal{L}^{-1}\{F(s)\,e^{-(x/c)s}\} = f(t-x/c)\,u(t-x/c)$$

In other words: $\omega(x,t) = \begin{cases} \sin(t - x/c) & \text{if } x/c < t < 2\pi + x/c \\ 0 & \text{otherwise} \end{cases}$

This is a single sine wave traveling to the right with speed c.

10.7